A large integral point on the cubic surface $x^2y + xy^2 + xz^2 + 2x^2 - z^2 + 3 = 0$

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Abstract

We report the integer solution

$$(x, y, z) = (-2450412151997178,\ 2357619934960084,\ 467727250324161)$$

of $x^2y + xy^2 + xz^2 + 2x^2 - z^2 + 3 = 0$, together with the three companions obtained from the symmetries $z \mapsto -z$ and $y \mapsto -x - y$. It was found by a complete, sieve-based search over x that ran for about two days on a 2-core cloud machine and an 8-core Apple M1 Pro. We explain the elementary reductions and the local–global arithmetic (including a Brauer–Manin type reciprocity condition) that confine x to $x \equiv 6 \pmod{72}$ and to a set of density about $1/1800$ in the integers, why one should expect the first solution to be large, why "small-parameter" methods (elliptic fibrations, caps) could not have found this particular point, and how the search was organised, validated and executed. The search was completed up to $|x| < 2.452 \cdot 10^{15}$. Up to the symmetries above, the point is the only solution in that range, and so the one with the smallest $|x|$ (Theorem 10).

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1 The result

Theorem 1. *The quadruples*

$$\begin{array}{rcl} x & = & -2450412151997178, \\ y & \in & \{\,2357619934960084,\ \ 92792217037094\,\}, \\ z & = & \pm 467727250324161 \end{array}$$

(four triples in total) satisfy

$$x^2y + xy^2 + xz^2 + 2x^2 - z^2 + 3 = 0. \tag{1}$$

Proof. Direct computation. For

$$(x, y, z) = (-2450412151997178,\ 2357619934960084,\ 467727250324161)$$

the six terms are:

x^2y	1415637537913250289658027536879244602806958945 6
xy^2	−1362030170043791981945251280379919710449592796 8
xz^2	−5360736786945948673984111800748382722332349 38
$2x^2$	12009039429310881955628239927368
$-z^2$	−218768780695800366279580353921
$+3$	3
sum	0

The left side of (1) is even in z, and it is invariant under $y \mapsto -x-y$ because $x^2y+xy^2 = xy(x+y)$ is. This gives $-x - y = 92792217037094$. □

The numbers have 15–16 decimal digits. The rest of this note explains why a solution of this size was to be expected, and how it was found.

2 First reductions

2.1 Parity

If x is odd then $xy(x + y)$ is even, $z^2(x - 1)$ is even and $2x^2$ is even, so the left side of (1) is odd. Hence x *is even*. Write

$$x = 2a, \qquad y = b - a \qquad (a, b \in \mathbb{Z}).$$

Since $x^2y + xy^2 = xy(x + y) = 2a(b - a)(b + a) = 2a(b^2 - a^2)$, equation (1) becomes

$$\boxed{2a\,b^2 + (2a-1)\,z^2 \;=\; N(a) := 2a^3 - 8a^2 - 3} \tag{2}$$

For the solution of Theorem 1,

$$a = -1225206075998589, \quad b = 1132413858961495, \quad z = 467727250324161.$$

2.2 The conic bundle and the circle at infinity

For fixed a, (2) is a conic in (b, z). For $a > 0$ both coefficients are positive. For $a < 0$ one multiplies by -1. Either way it is an *ellipse*, so each fibre carries finitely many integral points. Dividing by $2a$ gives

$$b^2 + z^2\Big(1 - \frac{1}{2a}\Big) = a^2 - 4a - \frac{3}{2a},$$

so asymptotically the point (b, z) lies on the circle $b^2 + z^2 \approx a^2$. It is natural to write

$$b \approx |a| \cos\varphi, \qquad z \approx |a| \sin\varphi.$$

The found solution has $\varphi = \arctan(z/b) \approx 0.3917$ rad (22.4°), with $b/|a| \approx 0.924$ and $z/|a| \approx 0.382$. This is a perfectly "generic" angle, a fact that matters in Section 4.4.

A second useful quantity is

$$k := \frac{z^2 - 3}{2a} \in \mathbb{Z}. \tag{3}$$

Indeed $z^2 \equiv 3 \pmod{2a}$ by (2) taken modulo $2a$. Substituting $z^2 = 2ak + 3$ into (2) gives the equivalent pair

$$z^2 = 2ak + 3, \qquad b^2 = (a-2)^2 - (2a-1)k - 7. \tag{4}$$

For our point $k = -89278360996331 = -13 \cdot 431 \cdot 7297 \cdot 2183641$.

3 Local conditions

Throughout, (x, y, z) is an integral solution, $x = 2a$, $y = b - a$, and $t := |2a - 1|$.

Lemma 2 (conditions at the primes dividing a)**.** *a is odd, z is odd, and $z^2 \equiv 3 \pmod{2a}$. Every prime $p \geq 5$ dividing a satisfies $p \equiv \pm 1 \pmod{12}$. Moreover $9 \nmid a$.*

Proof. Reduce (2) modulo $2a$: $-z^2 \equiv -3$. If a were even then $4 \mid 2a$ and $z^2 \equiv 3 \pmod 4$, which is impossible. So z is odd. For $p \geq 5$, $p \mid a$, 3 is a square mod p, i.e. $\left(\frac{3}{p}\right) = 1$, i.e. $p \equiv \pm 1 \pmod{12}$. If $9 \mid a$ then $3 \mid z$ and $z^2 \equiv 3 \pmod 9$, which is impossible. □

Lemma 3 (conditions at 2)**.** *$a \equiv 3 \pmod 4$ and b is odd.*

Proof. Since z is odd, $z^2 \equiv 1 \pmod 8$, and $8a^2 \equiv 0$. Reducing (2) modulo 8 gives $2ab^2 + 2a - 1 \equiv 2a^3 - 3$, i.e. $ab^2 \equiv a^3 - a - 1 \pmod 4$. As a is odd, $a^3 \equiv a \pmod 8$, so $ab^2 \equiv -1 \pmod 4$. This forces b odd and $a \equiv -1 \pmod 4$. □

Lemma 4 (conditions at the primes dividing $2a - 1$)**.** *$(2b)^2 \equiv -19 \pmod t$. Consequently every prime $p \mid t$ is either 19, occurring to the first power only, or satisfies $\left(\frac{p}{19}\right) = 1$.*

Proof. Modulo $2a-1$ we have $2a \equiv 1$, so (2) gives $b^2 \equiv N(a) = \frac{(2a)^3}{4} - 2(2a)^2 - 3 \equiv \frac{1}{4} - 2 - 3 = -\frac{19}{4}$. For $p \neq 19$, quadratic reciprocity gives $\left(\frac{-19}{p}\right) = \left(\frac{p}{19}\right)$ because $-19 \equiv 1 \pmod 4$. If $19^2 \mid t$, then $19 \mid 2b$ and $19^2 \mid (2b)^2 + 19$, which is impossible. □

Lemma 5 (the modulus 3)**.** *$a \not\equiv 2 \pmod 3$.*

Proof. Suppose $a \equiv 2 \pmod 3$. Then (2) modulo 3 reads $4b^2 + 3z^2 \equiv 16 - 32 - 3$, i.e. $b^2 \equiv 2 \pmod 3$, which is impossible. □

Proposition 6 (a reciprocity obstruction)**.** $a \equiv 3 \pmod{36}$*, i.e.* $x \equiv 6 \pmod{72}$*.*

Proof. Consider the quaternion algebra $(x, 3)$ evaluated at the solution, i.e. the Hilbert symbols $(x, 3)_v$ for all places v of $\mathbb{Q}$. By the product formula, $\prod_v (x, 3)_v = 1$.

Places $v = \infty$ and $p \nmid 6x$. Here $(x, 3)_v = 1$: at ∞ because $3 > 0$, and at p because both entries are units.

Primes $p \geq 5$ with $p \mid x$. Here $(x, 3)_p = \left(\frac{3}{p}\right)^{v_p(x)} = 1$ by Lemma 2.

Hence $(2a, 3)_2 = (2a, 3)_3$.

At 2: $(2, 3)_2 = (-1)^{(9-1)/8} = -1$ and $(a, 3)_2 = (-1)^{\frac{a-1}{2} \cdot \frac{3-1}{2}} = (-1)^{(a-1)/2} = -1$ by Lemma 3. So $(2a, 3)_2 = +1$.

At 3:

- If $3 \nmid a$, then $a \equiv 1 \pmod 3$ by Lemma 5, and $(2a, 3)_3 = \left(\frac{2a}{3}\right) = \left(\frac{2}{3}\right) = -1 \neq 1$. This is a contradiction, so $3 \mid a$.
- By Lemma 2, $a = 3a'$ with $3 \nmid a'$. Then $(2a, 3)_3 = (3, 3)_3 \, (2a', 3)_3 = \left(\frac{-1}{3}\right) \left(\frac{2a'}{3}\right) = \left(\frac{a'}{3}\right)$. This must equal 1, so $a' \equiv 1 \pmod 3$, i.e. $a \equiv 3 \pmod 9$.

Together with $a \equiv 3 \pmod 4$ this gives $a \equiv 3 \pmod{36}$. □

Remark 7. Proposition 6 is an instance of the (integral) Brauer–Manin obstruction. The class of $(x, 3)$ is unramified on the affine surface away from the boundary. Locally, $a \equiv 1 \pmod{12}$ *is* solvable at 2 and at 3 separately; it is only the global reciprocity $\prod_v (x, 3)_v = 1$ that kills it. A direct check of 2-adic and 3-adic solvability, carried out by brute force modulo 2^{10} and 3^6, shows that local conditions alone allow $a \equiv 3 \pmod 4$ and $a \equiv 1 \pmod 3$ or $a \equiv 3 \pmod 9$. The analogous algebra $(x - 1, -19)$, attached to the other degenerate fibre $2a = 1$, turns out to impose nothing new.

Lemma 8 (the cubic $N(a)$)**.** *Let p be a prime with $p \nmid 2a(2a-1)$ and $p \parallel N(a)$. Then* $\left(\frac{-2a(2a-1)}{p}\right) = 1$*.*

Proof. If $p \mid b$ and $p \mid z$ then $p^2 \mid N(a)$, contrary to assumption. If exactly one of b, z is divisible by p, then (2) modulo p forces $p \mid 2a$ or $p \mid 2a - 1$. So $p \nmid bz$ and $(z/b)^2 \equiv -2a/(2a - 1)$, and $-2a(2a - 1)$ is a square mod p. □

Remark 9. One checks that $\gcd(N(a), a) \mid 3$ and $\gcd(N(a), 2a - 1) \mid 19$. When $19 \mid 2a - 1$ there is a further 19-adic condition on a mod 19^2 (in effect $N(a)/19$ must be a square mod 19), which the search did not exploit; it would have saved about 1%. For $5 \leq p \leq 23$, $p \neq 19$, the conditions above agree exactly with brute-force p-adic solvability modulo p^3.

How restrictive is this?

Lemmas 2–8 and Proposition 6 leave

- a in one class modulo 36;
- $a/3$ composed of primes $\equiv \pm 1 \pmod{12}$, which is "half of the primes";
- $|2a - 1|$ composed of quadratic residues mod 19 (again half of the primes), plus possibly one factor 19;
- no small prime of $N(a)$ violating Lemma 8.

At $|a| \approx 10^{15}$ only about 2% of the integers $a \equiv 3 \pmod{36}$ (between 1.7% and 2.2% depending on the sign of a) survive all four conditions, i.e. about one integer in 1800.

The found point satisfies every condition

a	$-1225206075998589 \equiv 3 \pmod{36}$
$a/3$	$-23 \cdot 73 \cdot 457 \cdot 18803 \cdot 28307$ (residues mod 12 : $11, 1, 1, 11, 11$)
$2a-1$	$-40161731 \cdot 61013609$ (residues mod 19 : $6, 11$; both squares)
$N(a)$	$-3 \cdot 13063 \cdot 23567 \cdot 1928926393102203289 \cdot 2064777611218118087$
$\left(\frac{-2a(2a-1)}{p}\right)$	$+1$ for each of the four primes $p \neq 3$ of $N(a)$
$a \bmod 8,\ b \bmod 8,\ z \bmod 8$	$3,\ 7,\ 1$ $\quad (b, z \text{ odd};\ 3 \mid z)$

Two of the primes of $N(a)$ have 19 digits. No sieve could have certified the condition of Lemma 8 for them in advance; the search only ever checks it for $p < 1.5 \cdot 10^6$.

4 Why the first solution is large

4.1 Geometry

Homogenise (1) with w: $x^2y + xy^2 + xz^2 + 2x^2w - z^2w + 3w^3 = 0$. This is a smooth cubic surface $S \subset \mathbb{P}^3$. Its section at infinity $\{w = 0\}$ is

$$x\,(xy + y^2 + z^2) = 0,$$

the union of the line $L_\infty = \{x = w = 0\}$ and the conic $C_\infty = \{xy + y^2 + z^2 = 0 = w\}$. These meet in the two complex conjugate points $y = \pm iz$. So the affine surface $U = S \setminus (L_\infty \cup C_\infty)$ is a *log Calabi–Yau surface* whose boundary is a cycle of two rational curves. The real points of U go to infinity only along the real circle $C_\infty(\mathbb{R})$, which is the circle $b^2 + z^2 \approx a^2$ of §2.2. For such surfaces integral points are expected to be sparse, with counts growing only like a power of log of the height.

The only conic bundle defined over $\mathbb{Q}$ is the pencil of planes through L_∞, i.e. the x-fibration of §2.2, and its fibres are ellipses. There is therefore no Pell-type family of solutions; nor did a search find rational lines or twisted cubics on U that could parametrise solutions.

4.2 A probabilistic count

Fix an admissible a. The congruence $z^2 \equiv 3 \pmod{2a}$ has $2^{\omega(a/3)}$ roots modulo $|a|$, where $\omega(n)$ is the number of distinct prime factors of n. Each gives one candidate $z \in [0, |a|]$ up to sign, and then b^2 is determined by (4). A number of size about $a^2 \cos^2\varphi$ is a perfect square with probability about $1/(2|a\cos\varphi|)$. The local conditions modify this: for instance, each prime of $2a-1$ doubles the chance that b^2 is a square. Summing over admissible $|a| \le A$, the root counts 2^ω compensate exactly for the thinness of the admissible set, and one finds

$$\#\{\text{solutions with } |a| \le A\} \ \approx\ c \log A$$

with a small constant c. The constant was estimated numerically from the sibling equations $x^2y + xy^2 + xz^2 + 2x^2 - z^2 + c = 0$ ($c \neq 3$), which have many small solutions. For $c = 3$, the restriction $a \equiv 3 \pmod{36}$ and the two quadratic-character conditions make the constant much smaller: roughly 0.1–0.3 expected solutions per factor 10 in $|a|$.

The first solution can therefore easily lie at $|a| \approx 10^{15}$ or beyond, which is what happened. The expected number of solutions in any dyadic range $[A, 2A]$ is roughly constant, while the cost of searching it is proportional to A. This is the usual situation for log Calabi–Yau equations, the sums of three cubes being the best-known example.

4.3 Algebraic structure: two class-number-one rings

Two norm forms are hidden in (4).

The $\mathbb{Z}[\sqrt{3}]$ side. $z^2-3 = 2a\cdot k$, so $2a$ and k are complementary divisors of the norm $N_{\mathbb{Q}(\sqrt{3})/\mathbb{Q}}(z+\sqrt{3})$. In terms of binary quadratic forms, $[2a, 2z, k]$ has discriminant 12 and is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to $X^2 - 3Y^2$.

The $\mathbb{Z}[\omega]$ side, $\omega = \frac{1+\sqrt{-19}}{2}$ (class number 1). Completing the square in (4) gives

$$(a-k-2-b)(a-k-2+b) = k^2+3k+7 = N(k+1+\omega). \tag{5}$$

So with $d' = a-k-2-b$ and $e' = a-k-2+b$ we get $d'e' = k^2+3k+7$. Equivalently, $[e',\ 2k+3,\ d']$ is a binary quadratic form of discriminant -19, hence equivalent to $X^2+XY+5Y^2$. We have $2a-1 = e' + (2k+3) + d'$, and the whole equation collapses to

$$2z^2 = (T+1)(T-A-C-3)+6, \qquad A = e',\ C = d',\ T = A+B+C = 2a-1,\ B = 2k+3. \tag{6}$$

Here (A, C, T) are the values of $X^2+XY+5Y^2$ on the three regions around a vertex of Conway's topograph, and B is the label on the corresponding edge. Thus integral points of (1) correspond to vertices of the (negative-definite for $a < 0$) topograph at which one quadratic expression is twice a square.

For our point: $d' = -2268341573963755 = -5\cdot 541\cdot 624667\cdot 1342433$, $e' = -3513856040765 = -5\cdot 7\cdot 359\cdot 279654281$, $B^2-4AC = -19$, and (6) holds. Note also $d' = (y^2+y+5)/(x-1)$.

This structure explains Lemma 4, since $2a-1$ is a norm from $\mathbb{Z}[\omega]$, and Lemma 2, since $2a$ is a norm from $\mathbb{Z}[\sqrt{3}]$. It does not give a shortcut: walking the topograph costs more per useful candidate than the sieve described below.

4.4 Caps and elliptic fibrations

Fixing d' in (5) makes $z^2 = k(d'+e') + 2k^2+4k+3$, with $e' = (k^2+3k+7)/d'$, a genus-one curve. Explicitly it is

$$E_{d'}:\ W^2 = X^3 + d'(2d'+3)X^2 + d'^2(d'^2+4d'+7)X + 3d'^4, \qquad X = d'k,\ W = d'^2 z.$$

Similarly fixing $F = a+2k-2z$ (with $FG = s^2-12$, $s = a-2k$, $G = a+2k+2z$) gives another elliptic fibration. Solutions with small $|d'|$ live near $\varphi = 0$; solutions with small $|F|$ live near $\varphi = \pi/2$. These "caps" were searched extensively in exploratory computations: rank and generator computations for $|d'|, |F| \le 1000$, and divisor sieves for $|k|, |s| \le 10^9$ with $|d'|, |F| \le 10^6$. Nothing was found.

The point found here has $d' \approx -2.27\cdot 10^{15}$, $F = -2339217298639573$ and $k \approx -8.9\cdot 10^{13}$, all of the same order as a. It is a generic point in the middle of the surface, far from every cap. No small-parameter fibre contains it, so the only practical way to reach it was a complete search over a.

5 The search algorithm

The search enumerates $a = s\,m$ with $s = \pm 1$ and $m = r + 36j$, where $r = 3$ for $s = +1$ and $r = 33$ for $s = -1$, so that $a \equiv 3 \pmod{36}$ in both cases. It processes j in segments of 2^{19} consecutive values. The sizes involved are m up to $1.2\cdot 10^{15}$ and $|z| \le m+5$. For each segment:

1. **Exclusion sieve.** Mark j as impossible when:

- a prime $p \equiv \pm 5 \pmod{12}$ with $p \le \sqrt{m_{\max}}$ divides m (Lemma 2);
- a prime p with $\left(\frac{p}{19}\right) = -1$ and $p \le \sqrt{2m_{\max}}$ divides $t = 2m - s$ (Lemma 4);
- $361 \mid t$;
- $a \equiv \theta \pmod{p}$ for one of the 57,237 simple roots θ of N modulo primes $p < 1.5 \cdot 10^6$ with $\left(\frac{-2\theta(2\theta-1)}{p}\right) = -1$, unless a lies on the Hensel lift of θ modulo p^2 (Lemma 8).

The primes $5, 7, 13, 17, 19$ and $29, 31, 37$ are applied through two precomputed periodic byte patterns. Primes larger than the segment are handled by a bucket sieve, whose entries store the prime itself.

2. **The cofactor of t without factoring.** After sieving every bad prime up to $\sqrt{t}$, at most one prime of t is unknown. By multiplicativity of the Legendre symbol it is good if and only if t^*, which is t with one factor 19 removed, is a square modulo 19. One table lookup per survivor replaces a factorisation of t.

3. **Factor** $m/3$ over the good primes $p \equiv \pm 1 \pmod{12}$, $p \le \sqrt{m_{\max}}$, by a second sieve restricted to survivors (a bitmap with rank). The remaining cofactor q is 1 or a prime, automatically $\equiv \pm 1 \pmod{12}$.

4. **Square roots of 3 modulo** q, batched four at a time with Montgomery arithmetic and lazy reduction:

   - $q \equiv 3 \pmod 4$: $\rho = 3^{(q+1)/4}$;
   - $q \equiv 5 \pmod 8$: Atkin's formula with base 6;
   - $q \equiv 1 \pmod 8$: Tonelli–Shanks, with a non-residue found from quadratic reciprocity by table lookups.

   The square roots of 3 modulo the small primes are precomputed and lifted by Hensel's lemma for prime powers.

5. **CRT with sign symmetry.** All roots of $z^2 \equiv 3 \pmod m$ are combined from the prime roots, with $z \equiv 0 \pmod 3$. Only one representative of each pair $\{R, m - R\}$ is built. Both odd lifts z and $2m - z$ in $[0, m + 5]$ are then tested.

6. **Candidate tests.** Compute $k = (z^2 - 3)/(2a)$ exactly, using an inverse modulo 2^{64} and verifying the result, and

$$B_2 = (a - 2)^2 - (2a - 1)k - 7 = m^2 - 4sm - z^2 + sk.$$

   Then apply, in order:

   - $B_2 > 0$;
   - $B_2 \equiv 1 \pmod 8$ (Lemma 3);
   - B_2 is a square modulo $45045 = 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13$ and modulo $7429 = 17 \cdot 19 \cdot 23$, using signed residue arithmetic;
   - an exact integer square root.

   Only about 1.2% of the candidates reach the last step.

Typical statistics for a $2 \cdot 10^{11}$-wide chunk of $|a|$ near 10^{15}, for one sign, are $5.6 \cdot 10^9$ values of j, of which $1.2 \cdot 10^8$ survive the sieves. These produce $3.1 \cdot 10^8$ candidates z, of which $3.7 \cdot 10^6$ need an exact square root. The cost is linear in the range of $|a|$ searched. Throughput was about $2 \cdot 10^9$ values of $|a|$ per second (both signs) on two Intel Xeon (Cascade Lake, 2.8 GHz) cores, and about $9 \cdot 10^9$ per second with eight worker processes on an Apple M1 Pro.

6 Validation, and a bug worth recording

An independent implementation. A short Python/SymPy program, `check.py`, re-derives everything from first principles:

- it factors $a/3$ and $2a - 1$;
- it tests the N-condition by trial division over all primes below the bound;
- it computes all roots of $z^2 \equiv 3 \pmod{2|a|}$ with `sqrt_mod`;
- it applies the same candidate tests.

On windows at $|a| \approx 10^5$, $3.6 \cdot 10^7$, $3.6 \cdot 10^{10}$ and 10^{13}, for both signs, the C program and `check.py` agree exactly. They agree on the counts of kept a, of candidates and of survivors of each filter, and on the full list of (m, z) pairs reaching the exact square-root test. Every later optimisation was required to reproduce these lists exactly. These were bucket sieving, SIMD scans, lazy Montgomery arithmetic, the portable Mac build and L1 blocking. The Mac build also ran a self-test of six known windows before starting.

The bug. An earlier generation of the program computed the residue of B_2 modulo 45045 and 7429 with an expression that mixed `uint64_t` and `int64_t`. C's usual arithmetic conversions then evaluated it in unsigned arithmetic. Whenever the intermediate value was negative, the residue was taken of $2^{64} + v$ instead of v, and a genuine square was rejected with high probability. Because all versions shared the line, comparing versions against each other could not reveal it. It surfaced when a rewrite produced twice as many square-root candidates. The effective coverage of that first run, up to $|a| \approx 10^{13}$, was only about 30%. The whole range was re-searched from zero with the corrected code, which is the version validated against `check.py` as above.

Sanity checks inside the program. The program aborts if any of the following fails:

- every root satisfies $z^2 \equiv 3 \pmod{m}$, checked through the exactness of k;
- the CRT modulus equals m;
- the large cofactor is $\equiv \pm 1 \pmod{12}$.

A wrong root therefore cannot silently lose a candidate.

7 The computation

date and time (CET/CEST)	event	coverage
24 Oct, 00:10	corrected search restarted from $a = 0$ (cloud, 2 cores)	
24 Oct, 01:48	8 workers started on the M1 Pro	
24 Oct, 03:17	all $\lvert a\rvert < 2 \cdot 10^{13}$ complete	$\lvert x\rvert < 4 \cdot 10^{13}$
24 Oct, 03:17 onwards	interleaved chunks of width $2 \cdot 10^{11}$	Mac 5/6, cloud 1/6
25 Oct, 10:03 CET	**solution** found by Mac worker 2	chunk $\lvert a\rvert \in [1.2252, 1.2254) \cdot 10^{15}$

The solution came from the $s = -1$ half of the chunk, at $j = 34033502111071$, i.e. $m = 33 + 36j$ and $a = -m$. For this a there were $2^5 = 32$ roots of $z^2 \equiv 3 \pmod{|a|}$, giving 16 candidates up to sign, and exactly one of them made B_2 a perfect square. When the solution appeared, the workers stopped by themselves.

Searched range. When the solution appeared, the interleaved layout had left a few chunks below it unfinished. These were the cloud's share between $1.18 \cdot 10^{15}$ and $1.2254 \cdot 10^{15}$, plus one interrupted half-chunk: 20 intervals for $s = +1$ and 21 for $s = -1$, each $5.6 \cdot 10^9$ values of j wide. They were searched afterwards with the same program. None of the 41 gap runs found a solution. The searched range is therefore complete for both signs of a:

$$|a| < 1.226 \cdot 10^{15}, \qquad \text{i.e.} \qquad |x| < 2.452 \cdot 10^{15}.$$

It was covered by 12,425 completed program runs, with every j-interval accounted for in the logs. In this whole range the only line ever reported was the one solution above.

Theorem 10 (computational)**.** *Up to the symmetries $z \mapsto -z$ and $y \mapsto -x - y$, the solution of Theorem 1 is the only integral solution of* (1) *with $|x| < 2.452 \cdot 10^{15}$. In particular it is the solution with the smallest $|x|$.*

This statement rests on two things: the proofs in Section 3, which justify every exclusion the program makes, and the correctness of the program. The program was validated as described in the previous section.

8 Remarks

- The heuristic suggests infinitely many solutions, with about $c \log A$ of them up to height A. The next one is expected to be much further out, by one or more factors of 10 in $|a|$.
- The method is a direct analogue of Booker–Sutherland-style searches for sums of three cubes, but here every natural factorisation is balanced. The pairs $2a \cdot k$, $(2a-1) \cdot v$ with $v = 2a - 4k - 7$, $d' \cdot e'$ and $F \cdot G$ all have factors of the same size for generic points. So there is no "small factor" to exploit, and the search must run over a itself.
- The reciprocity argument of Proposition 6 cuts the residue classes of a modulo 36 that local conditions allow from four to one. More refined conditions from the Brauer group, if there are any, could reduce it further.

A Reproducing the verification

```
x, y, z = -2450412151997178, 2357619934960084, 467727250324161
f = lambda x, y, z: x*x*y + x*y*y + x*z*z + 2*x*x - z*z + 3
for Y in (y, -x - y):
    for Z in (z, -z):
        assert f(x, Y, Z) == 0
```

B Files

`search_mac.c`	the program that found the solution (portable C, single thread)
`run.sh`, `worker.sh`, `workers.sh`, `config.sh`	8-worker driver with atomic chunk claims
`selftest.sh`, `status.sh`, `stop.sh`	self-test, progress, graceful stop
`search14.c`, `drive_cloud.sh`	x86 version (OpenMP, AVX2) and cloud driver
`make_nbad.py`, `nbad.txt`	table of excluding roots of N modulo $p < 1.5 \cdot 10^6$
`check.py`	independent Python/SymPy re-implementation used for validation